

Illinois Institute of Technology
MMAE 350 – Computational Mechanics
Homework 6: 3D Printer Nozzle Cooling

Problem Description

During 3D printing, the brass nozzle tip reaches high temperatures. After printing stops, the tip cools due to conduction along the nozzle and convection to the surrounding air.

We model the nozzle as a one-dimensional rod of length

$$L = 0.020 \text{ m.}$$

Temperature $T(x, t)$ satisfies the transient heat equation:

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}, \quad 0 < x < L.$$

Material properties (brass):

$$k = 110 \text{ W/(m}\cdot\text{K)}, \quad \rho = 8500 \text{ kg/m}^3, \quad c_p = 380 \text{ J/(kg}\cdot\text{K)},$$

$$\frac{k}{\rho c_p}.$$

Initial and boundary conditions:

$$T(x, 0) = 220^\circ\text{C},$$

$$T(0, t) = 220^\circ\text{C} \quad (\text{heater block, Dirichlet}),$$

$$-k \frac{\partial T}{\partial x}(L, t) = h (T(L, t) - T_\infty) \quad (\text{convection at tip}),$$

where

$$h = 60 \text{ W/(m}^2\text{K)}, \quad T_\infty = 25^\circ\text{C}.$$

Part (a) – Derivation

Using a three-term Taylor series about $x_N = L$:

$$T_{N-1} = T_N - \Delta x T'_N + \frac{\Delta x^2}{2} T''_N,$$

derive an expression for T''_N in terms of T_{N-1} , T_N , and T'_N .

Then:

- (i) Use the Robin boundary condition to eliminate T'_N .
- (ii) Substitute into the heat equation.

(iii) Derive the explicit FTCS time update for T_N^{n+1} .

(iv) Define the discrete Biot number

$$\text{Bi} = \frac{h\Delta x}{k}.$$

Your final expression should be written in terms of T_{N-1}^n , T_N^n , Bi, T_∞ , and

$$\lambda = \frac{\Delta t}{\Delta x^2}.$$

Part (b) – Numerical Implementation

1. Discretize the domain using $N = 40$ intervals.

2. Choose a stable time step satisfying

$$\lambda \leq \frac{1}{2}.$$

3. March forward in time until $t = 60$ seconds.

4. Implement:

- Interior FTCS update
- Dirichlet boundary at $x = 0$
- Robin boundary at $x = L$

Part (c) – Plots

Produce:

1. Temperature profiles at

$$t = 0, 5, 15, 30, 60 \text{ s.}$$

2. A plot of the tip temperature $T(L, t)$ versus time.

All plots must include labeled axes and units.

Part (d) – Engineering Interpretation

Answer briefly:

1. Approximately how long does it take the tip to cool below 180°C ?
2. Repeat the simulation with $h = 10 \text{ W}/(\text{m}^2\text{K})$. How does cooldown time change?
3. Physically explain how increasing h affects the boundary update equation.
4. What physical condition does $h \rightarrow 0$ represent?

Submission Requirements

Submit:

- A PDF containing:
 - Your derivation (Part a)
 - Short-answer responses
- Your Jupyter notebook file (.ipynb) containing all required runs and plots